

Graded Lab2 Tasks 1 – Python Basics for AI

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Allotted Time: 1.5 Hours

Instructions

- Attempt all lab tasks individually.
- Use Python environment to implement the tasks.
- Submit your solutions in pdf format on the CUI portal and GitHub.

1 Task 1: Python Classes and Memory Concepts

Objective

This task focuses on understanding Python classes, attributes, instances, and memory usage concepts relevant to Artificial Intelligence.

Task Description

1. Create a Python program to instantiate a specified class and display the namespace of that instance.
2. Define a function **student_data()** that prints the student ID. Extend the function to optionally display student name and class when provided.
3. Create a class named **Student** and examine its type. Display the keys of its **__dict__** attribute and the value of its **__module__**.
4. Create two empty classes **Student** and **Marks**. Instantiate objects and verify:
 - Whether the objects belong to their respective classes
 - Whether these classes are derived from the built-in **object** class
5. Define a class **Student** with attributes **student_name** and **marks**. Modify attribute values and display both original and updated values.
6. Create a class **Student** with attributes **student_id** and **student_name**. Add a new attribute **student_class**, display all attributes, then remove **student_name** and display the updated attributes.

7. Extend the **Student** class by adding **student_class** and implement a function to display all attributes and their values.
8. Create two instances **student1** and **student2** of the **Student** class. Assign values to their attributes and display all attributes in a structured format.
9. Perform memory allocation analysis on the created Python class.
10. Apply techniques to reduce memory usage and observe the differences.

Lab Task 2: Object-Oriented Design and Area Comparison

Objective

This task develops understanding of Object-Oriented Programming and its application in solving mathematical problems and comparing results.

Task Description

1. Design two classes representing **Trapezoid** and **Parallelogram**. Define appropriate attributes and compute their areas using standard mathematical formulas.
2. Develop a comparison mechanism to evaluate areas of different shapes.
3. Store the computed areas in a dictionary structure.
4. Iterate over multiple sets of dimensions to analyze variations in area.
5. Utilize dictionary operations to determine and display the shape with the maximum area.